

Security Testing Report

SentinelLite: A Lightweight SIEM with Rule-Based Detection and ML Anomaly Scoring

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Test Dataset

The safe sample logs include Apache access logs, Linux auth logs, firewall syslog lines, and Windows Event CSV rows. All IP addresses use documentation-only ranges: 192.0.2.0/24, 198.51.100.0/24, and 203.0.113.0/24.

Detection Results

Alert Distribution

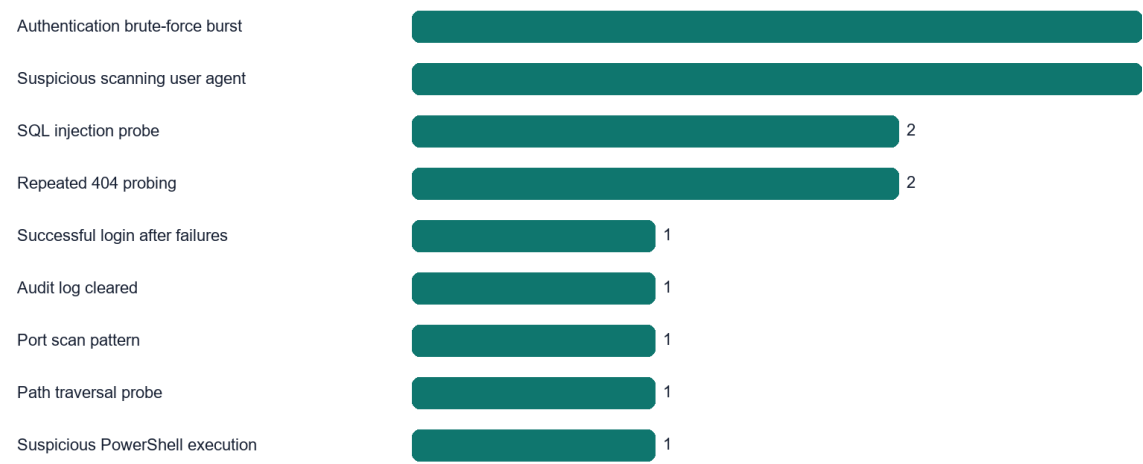


Fig. 3. Distribution of generated rule-based alerts.

Metric	Value
Parsed log events	49
Rule alert groups	15
Anomaly windows	3
Precision	1.0
Recall	1.0
F1 score	1.0

Expected Alert Evaluation

Alert	Source	Expected	Actual	Status	
SQL injection probe	203.0.113.45	1	1	1	pass
SQL injection probe	198.51.100.77	1	1	1	pass
Path traversal probe	203.0.113.45	1	1	1	pass
Suspicious scanning user agent	203.0.113.45	1	1	1	pass

Alert	Source	Expected	Actual	Status	
Suspicious scanning user agent	203.0.113.88	1	1	1	pass
Suspicious scanning user agent	198.51.100.77	1	1	1	pass
Repeated 404 probing	198.51.100.23	1	1	1	pass
Repeated 404 probing	203.0.113.88	1	1	1	pass
Authentication brute-force burst	198.51.100.99	1	1	1	pass
Authentication brute-force burst	203.0.113.70	1	1	1	pass
Authentication brute-force burst	198.51.100.150	1	1	1	pass
Successful login after failures	203.0.113.70	1	1	1	pass
Suspicious PowerShell execution	192.0.2.10	1	1	1	pass
Audit log cleared	192.0.2.20	1	1	1	pass
Port scan pattern	198.51.100.200	1	1	1	pass

Limitations

The labeled evaluation is intentionally small and safe. It proves functional correctness for the included demonstration data, but it does not prove production-grade performance on large enterprise traffic. Future work should test with public datasets such as HDFS, BGL, CICIDS, and live lab-generated logs.